Calibration results
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Normalized Residuals

Reprojection error (cam0):   mean 0.04673040186180495, median 0.04408582032125044, std: 0.024193687399715884
Reprojection error (cam1):   mean 0.04914330915829306, median 0.0459362285890967, std: 0.026177682522496342
Gyroscope error (imu0):   mean 0.09074487075888034, median 0.08522202683023265, std: 0.0442957704866094
Accelerometer error (imu0):   mean 0.0730806052328536, median 0.04988224494517616, std: 0.06123361670097268

Residuals

Reprojection error (cam0) [px]:   mean 0.04673040186180495, median 0.04408582032125044, std: 0.024193687399715884
Reprojection error (cam1) [px]:   mean 0.04914330915829306, median 0.0459362285890967, std: 0.026177682522496342
Gyroscope error (imu0) [rad/s]:   mean 0.004458336858198561, median 0.004186997018896861, std: 0.002176271392214514
Accelerometer error (imu0) [m/s^2]: mean 0.018493166857639458, median 0.012622783788742913, std: 0.015495267004695666

Transformation (cam0):

T_ci: (imu0 to cam0):
```
[[ 0.99993585  0.01025428  0.00480996  0.00645538]
 [-0.0102486   0.99994676 -0.00120429 -0.00402044]
 [-0.00482206  0.00115492  0.99998771 -0.01767134]
 [ 0.          0.          0.          1.        ]]
```

T_ic: (cam0 to imu0):
```
[[ 0.99993585 -0.0102486  -0.00482206 -0.00658138]
 [ 0.01025428  0.99994676  0.00115492  0.00397444]
 [ 0.00480996 -0.00120429  0.99998771  0.01763524]
 [ 0.          0.          0.          1.        ]]
```

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.0036244215341242184

Transformation (cam1):

```
-----------------------
T_ci:  (imu0 to cam1):
[[ 0.99993166  0.01023062  0.00565813 -0.04370921]
 [-0.01022672  0.99994745 -0.00071867 -0.00403598]
 [-0.00566518  0.00066075  0.99998373 -0.01768961]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.99993166 -0.01022672 -0.00566518  0.04356473]
 [ 0.01023062  0.99994745  0.00066075  0.00449463]
 [ 0.00565813 -0.00071867  0.99998373  0.01793373]
 [ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.0036165511674700218

Baselines:
----------
Baseline (cam0 to cam1):
[[ 0.99999964 -0.00002463  0.00084814 -0.05014969]
 [ 0.00002422  0.99999988  0.00048551 -0.00000712]
 [-0.00084816 -0.00048549  0.99999952 -0.00001475]
 [ 0.          0.          0.          1.        ]]
baseline norm:  0.05014969742550675 [m]


Gravity vector in target coords: [m/s^2]
[ 0.03434215 -9.71711442 -1.32095832]


Calibration configuration
=========================

cam0
-----
```

```
  Camera model: pinhole
  Focal length: [390.76399129477807, 391.20099803868953]
  Principal point: [318.0142217795937, 242.2784877782992]
  Distortion model: radtan
  Distortion coefficients: [0.004090083315162736, -0.004089996633748525, 0.0001978156144338315, 
0.0007481304721465267]
  Type: aprilgrid
  Tags: 
    Rows: 6
    Cols: 6
    Size: 0.088 [m]
    Spacing 0.026399999999999996 [m]

cam1
-----
  Camera model: pinhole
  Focal length: [390.6811721023493, 391.1079246042584]
  Principal point: [318.4564581234082, 242.43891273296373]
  Distortion model: radtan
  Distortion coefficients: [0.004913676723961613, -0.005459907346999315, 0.00010386970848960114, 
0.00020107428856592865]
  Type: aprilgrid
  Tags: 
    Rows: 6
    Cols: 6
    Size: 0.088 [m]
    Spacing 0.026399999999999996 [m]



IMU configuration
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IMU0:
 ----------------------------
  Model: calibrated
  Update rate: 200.0
```
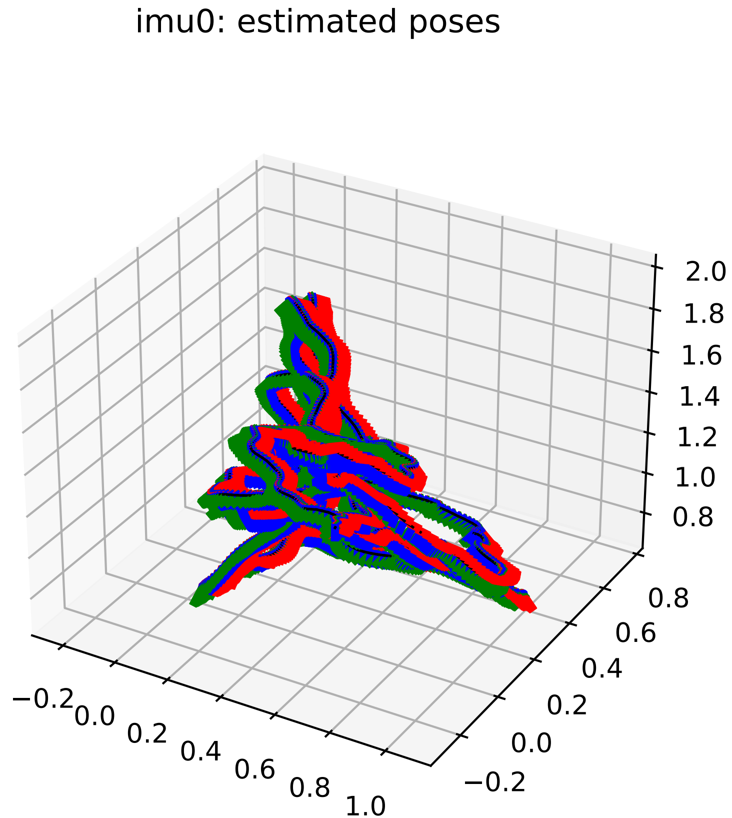
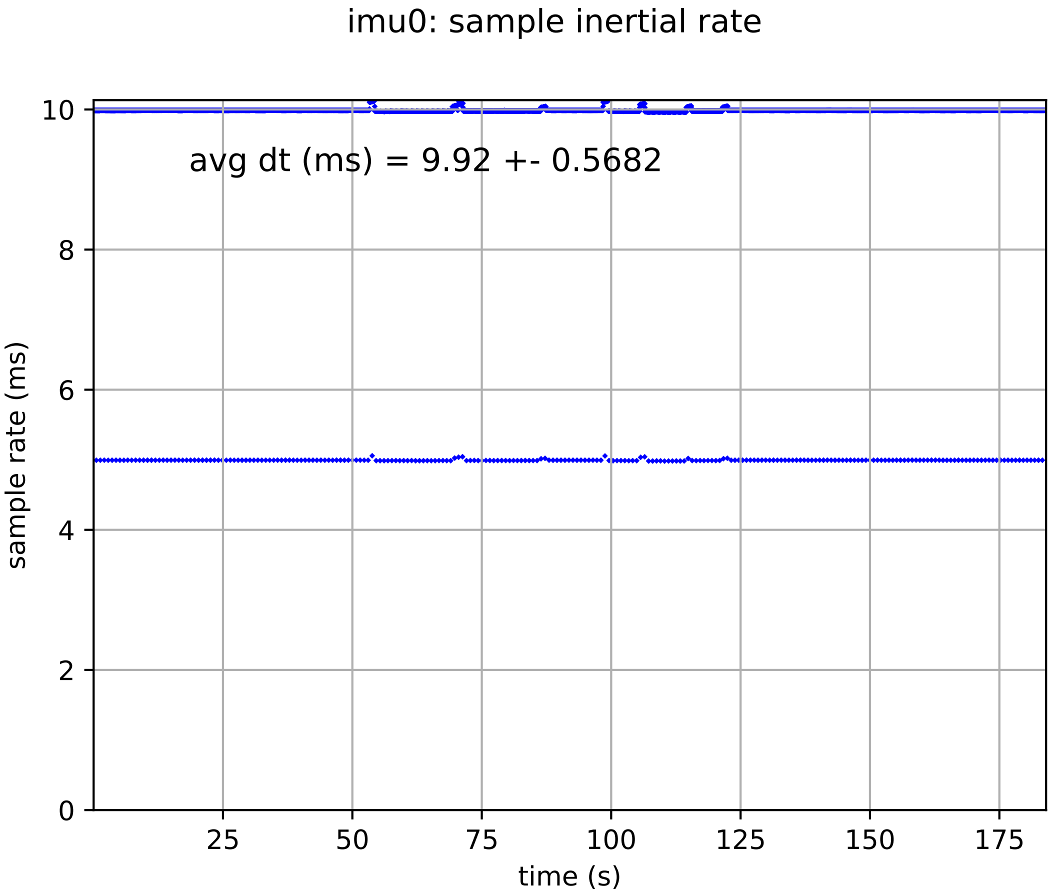
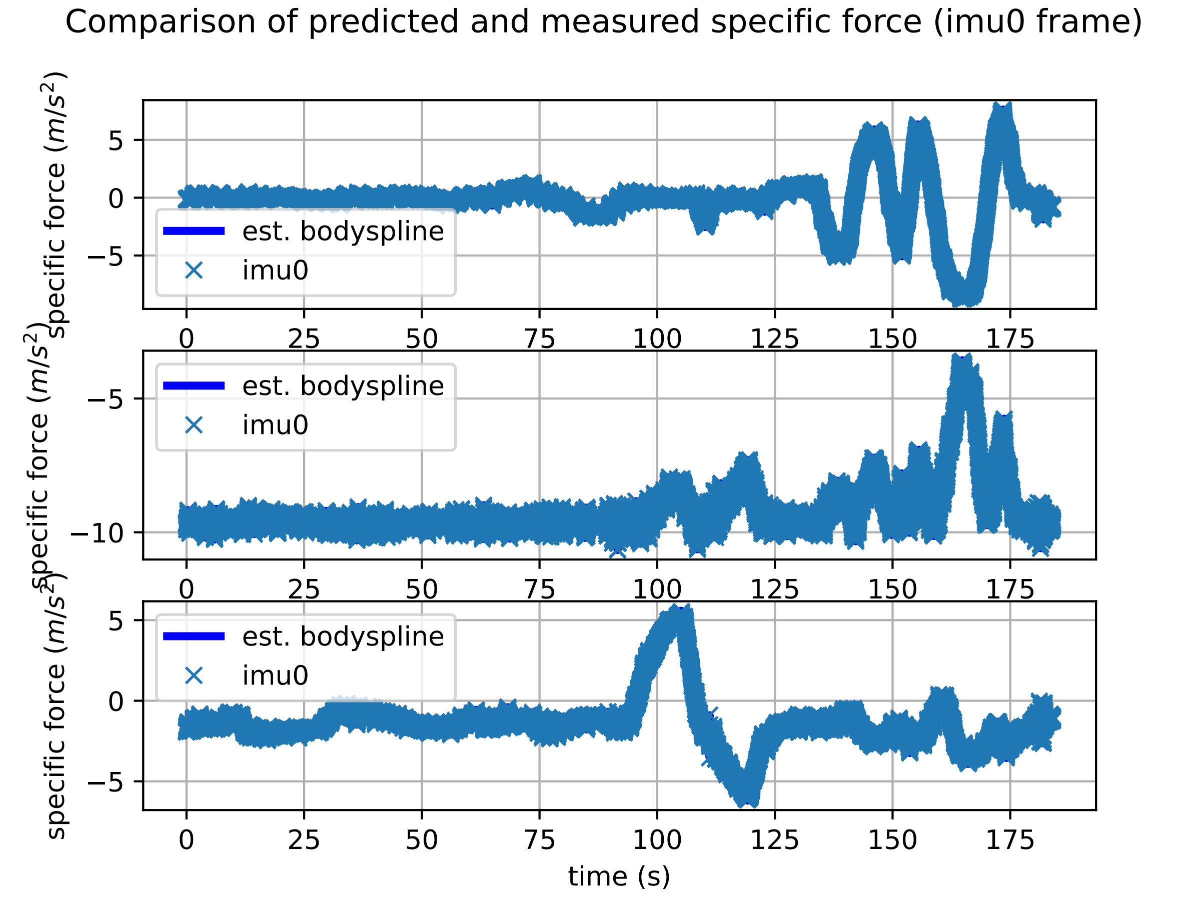

```
Accelerometer:
  Noise density: 0.017893452919534023
  Noise density (discrete): 0.25305163796489466
  Random walk: 0.00029022924848606
Gyroscope:
  Noise density: 0.003474047843015548
  Noise density (discrete): 0.04913045575925585
  Random walk: 2.106476770066359e-05
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
```

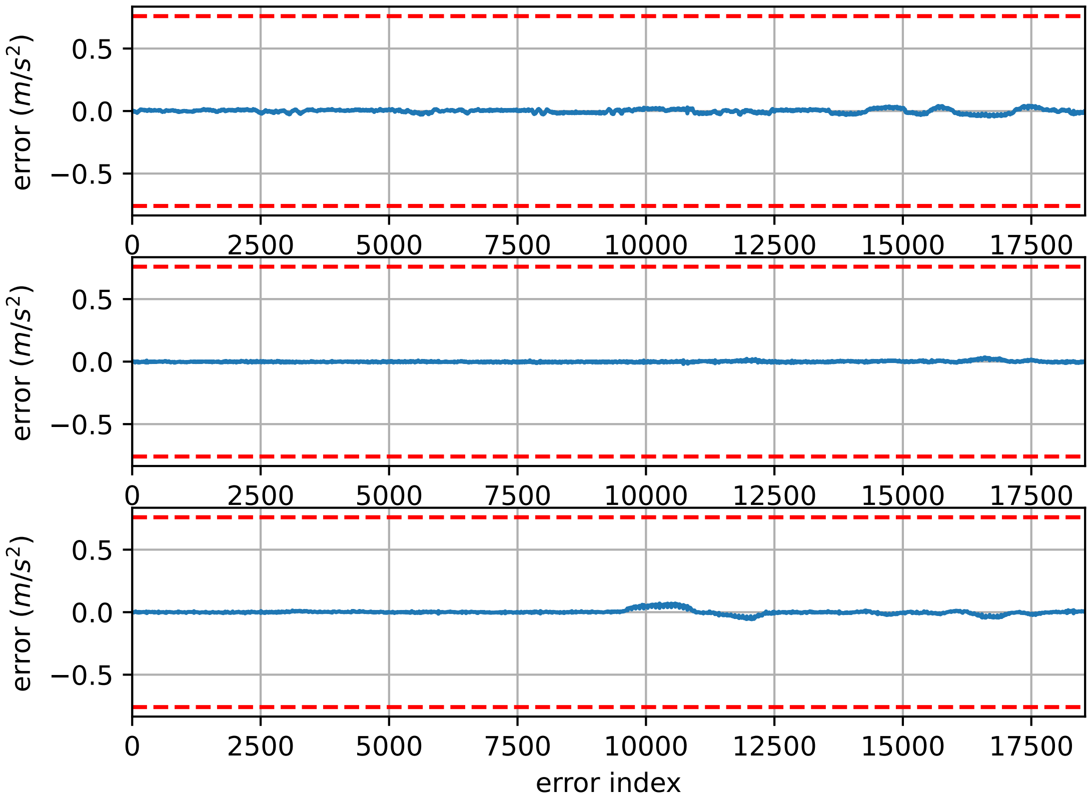

imu0: estimated poses
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.8
0.6
0.4
0.2
0.0
−0.2
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

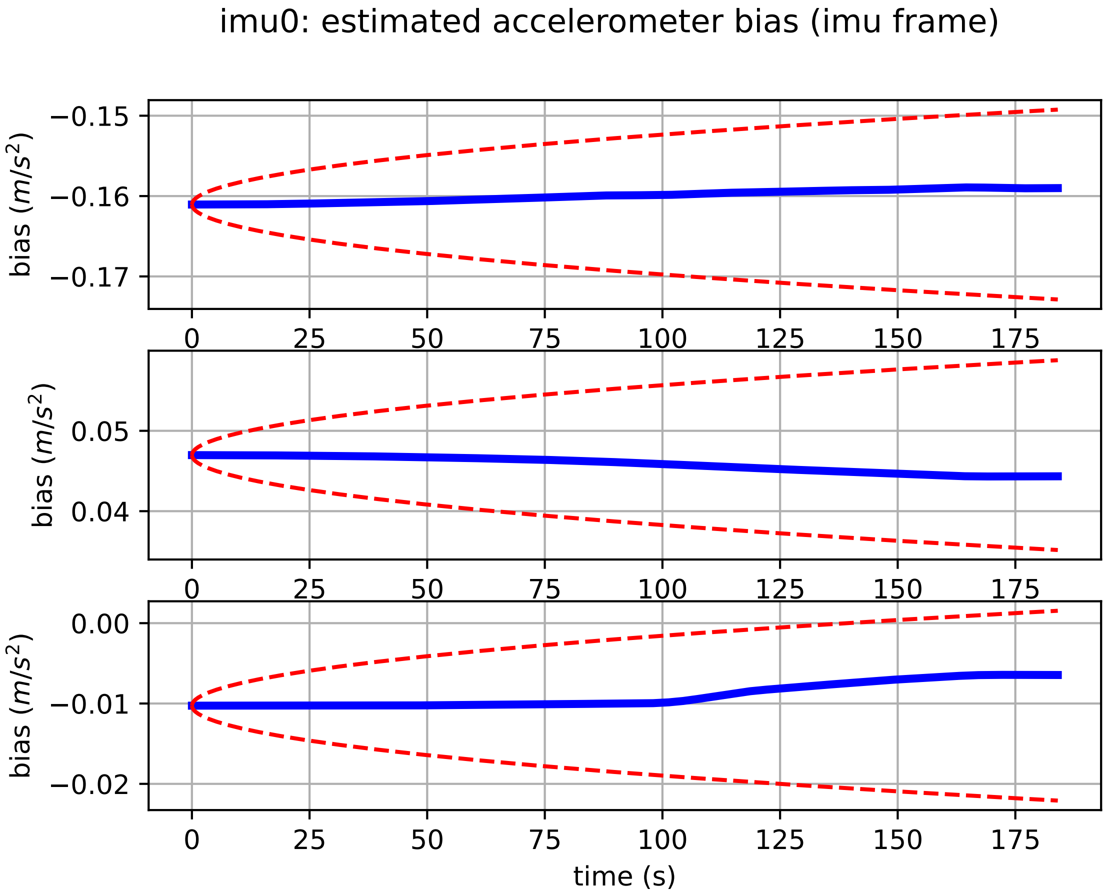


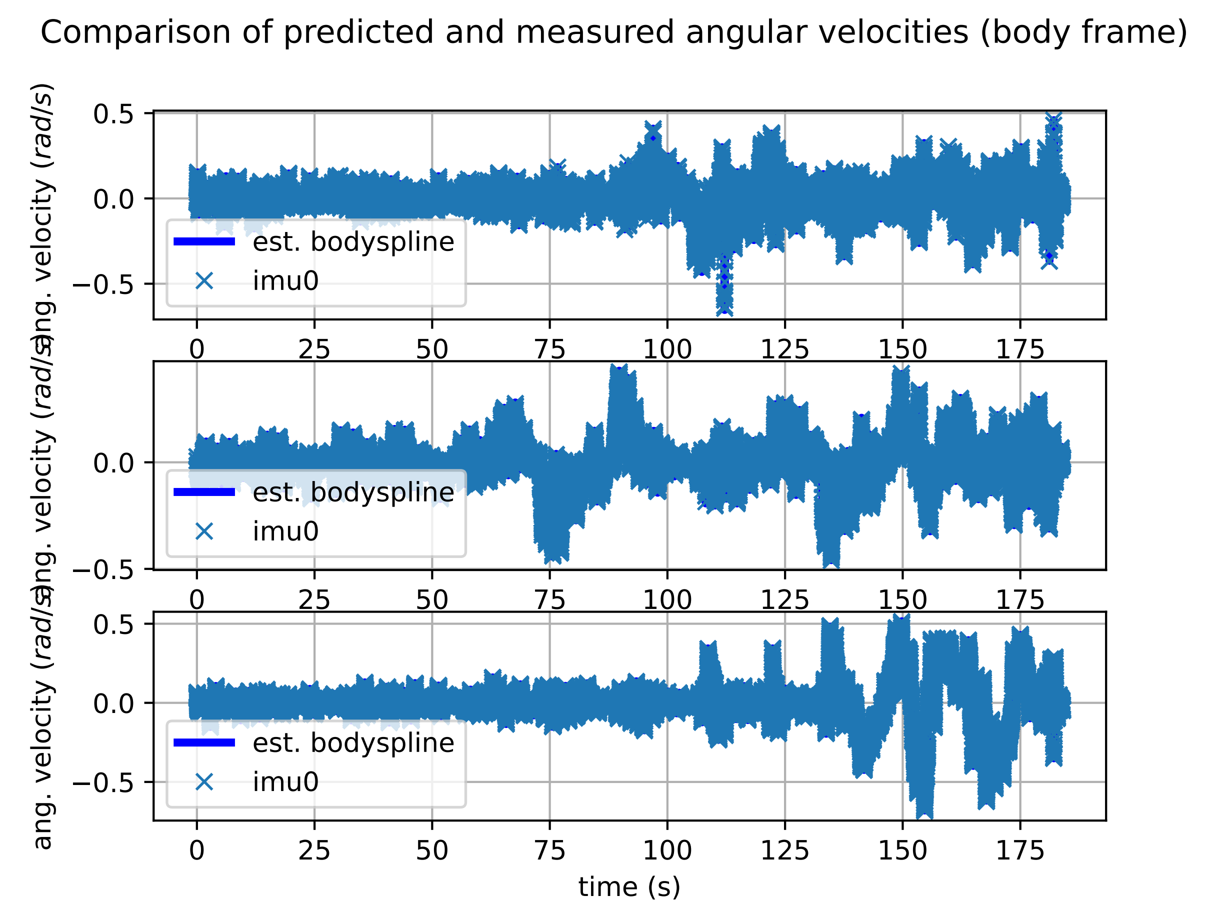
imu0: sample inertial rate
avg dt (ms) = 9.92 +- 0.5682
sample rate (ms)
time (s)
10
8
6
4
2
0
25
50
75
100
125
150
175


Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
time (s)
0
25
50
75
100
125
150
175
5
0
−5
−10

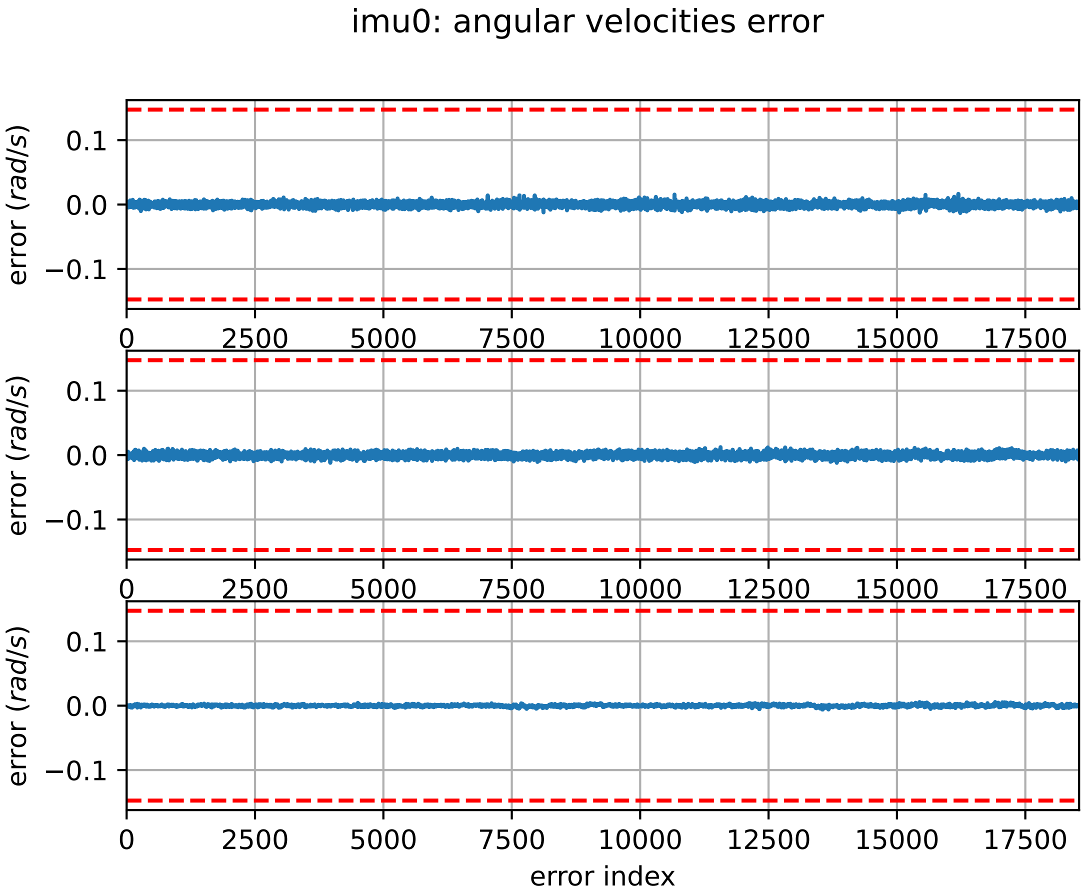
imu0: acceleration error

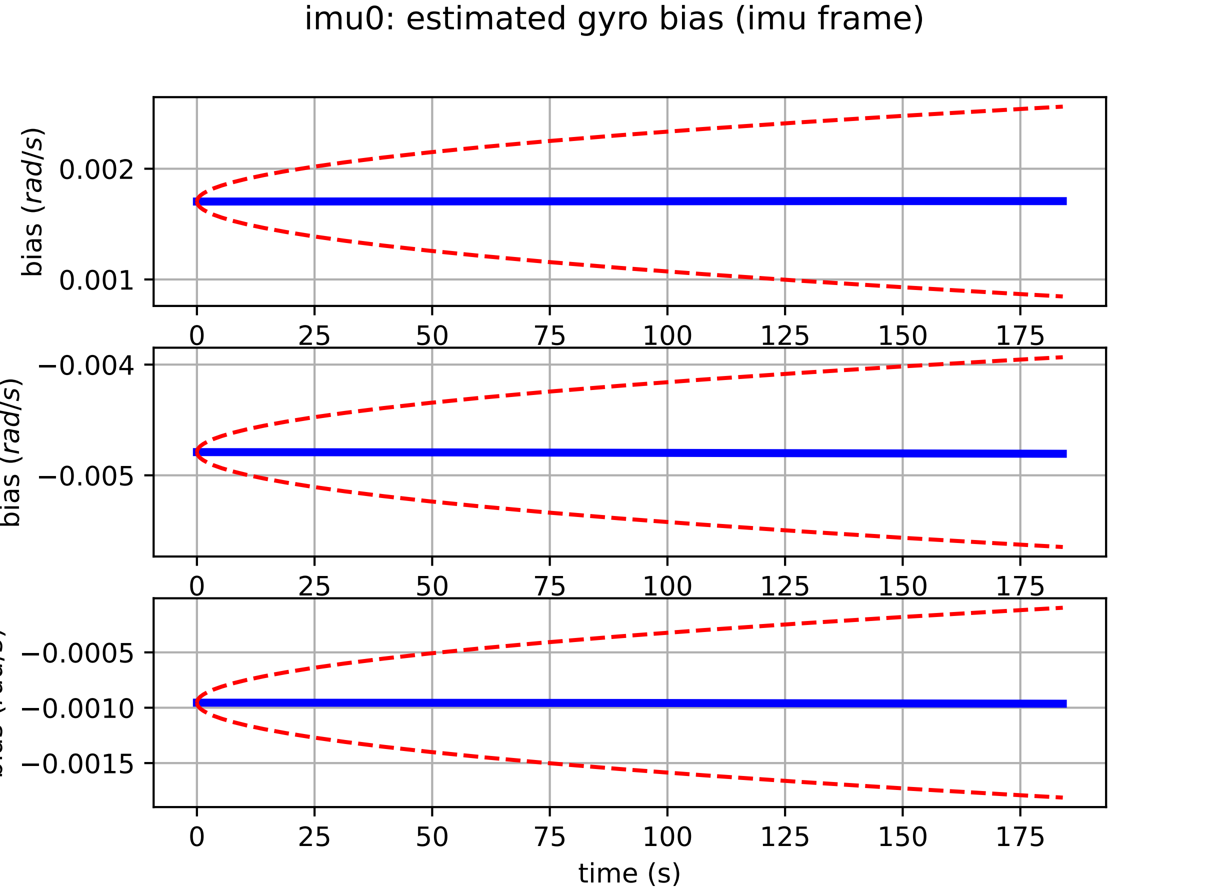



imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
−0.15
−0.16
−0.17
0.05
0.04
0.00
−0.01
−0.02
0
25
50
75
100
125
150
175
time (s)

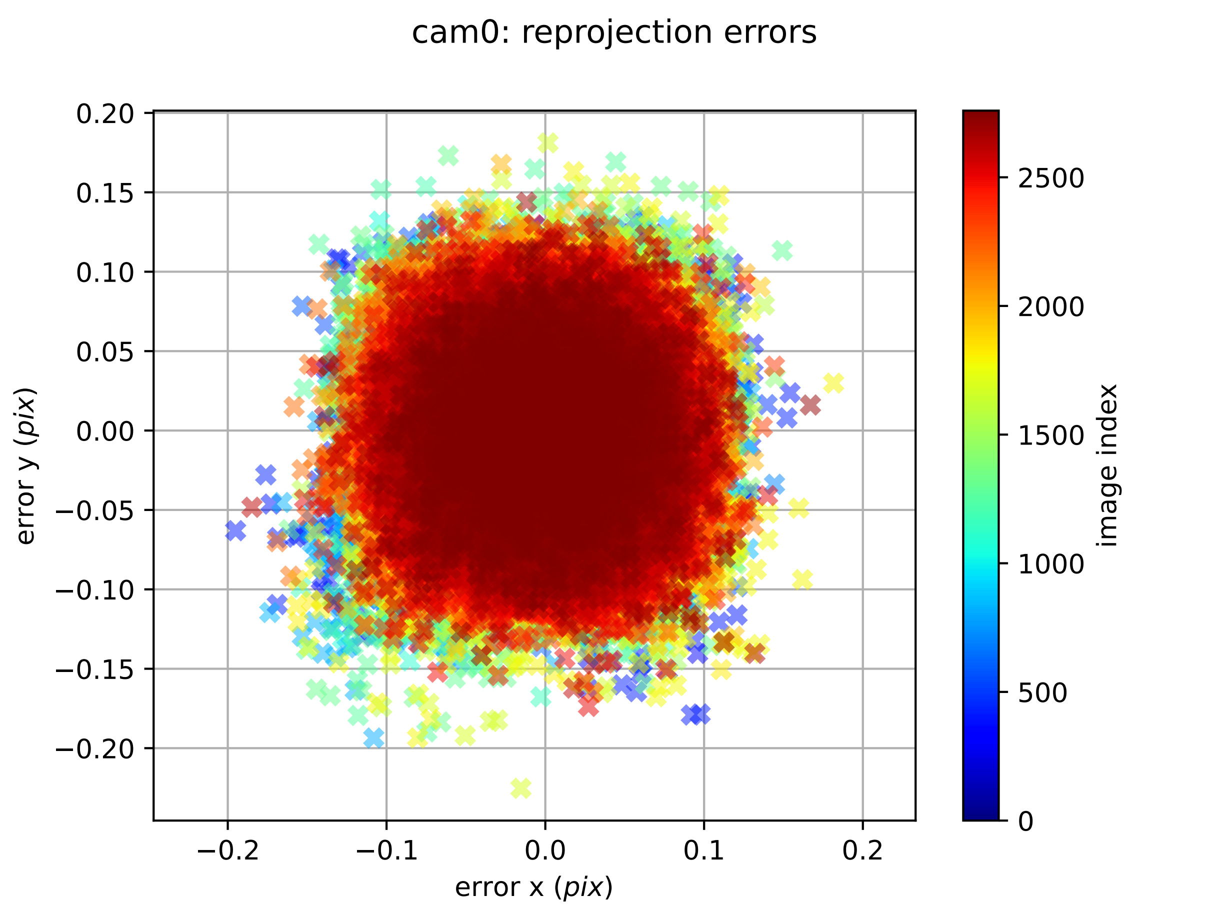


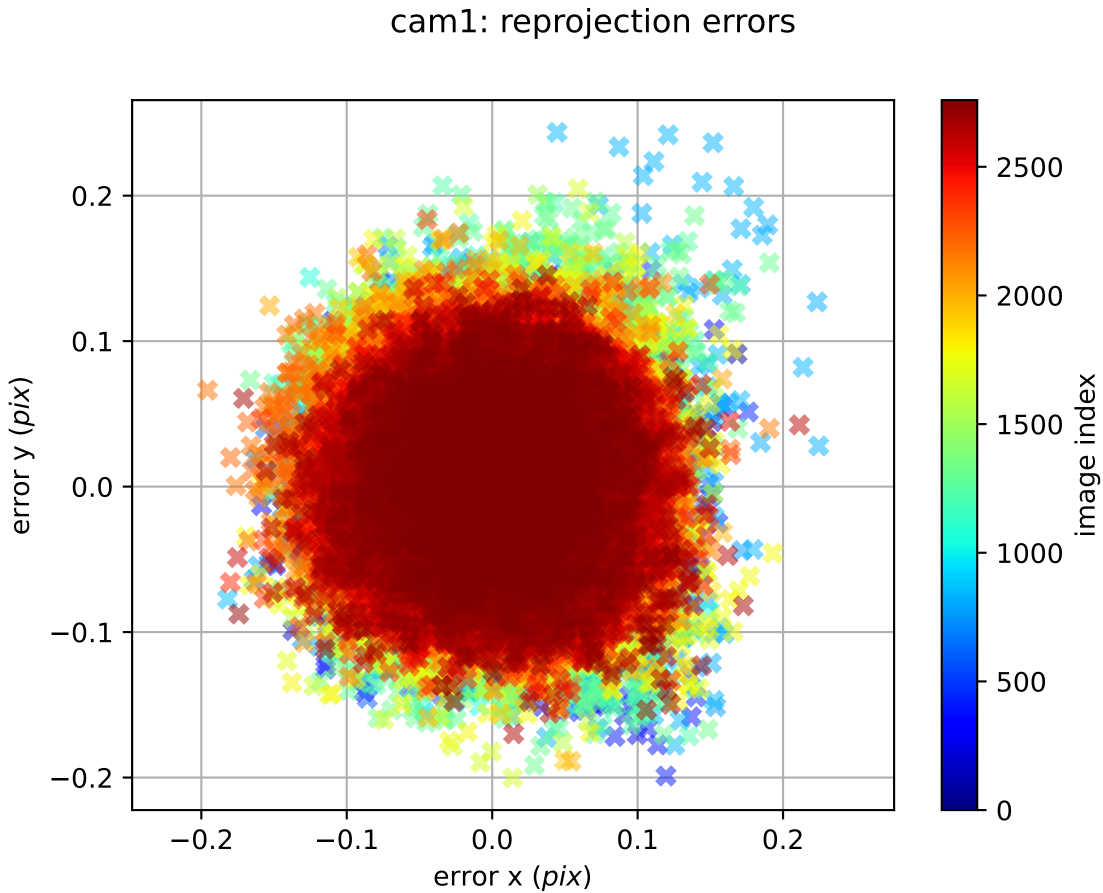
Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
0.5
0.0
−0.5
0
25
50
75
100
125
150
175
time (s)


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
2500
5000
7500
10000
12500
15000
17500
error index



imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.002
0.001
−0.004
−0.005
−0.0005
−0.0010
−0.0015
0
25
50
75
100
125
150
175
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.2
−0.1
0.0
0.1
0.2
2500
2000
1500
1000
500
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
0.2
0.1
0.0
−0.1
−0.2
2500
2000
1500
1000
500
0